PROJECT : HUMAN DETECTION AND SURVEILLANCE SYSTEM

1. Executive Summary

This project demonstrates the implementation of a real-time **Computer Vision (CV) Pipeline** for automated human presence identification and demographic analysis. By integrating **HOG (Histogram of Oriented Gradients)** for person detection and **Deep Neural Networks (DNN)** for age and gender classification, the system achieves autonomous situational awareness. This project confirms the viability of deploying multi-model AI solutions on local hardware to monitor environments without relying on cloud-based processing.

2. Infrastructure & Environment

The system was developed and tested in a localized environment to ensure low latency and data privacy.

- **Host OS:** Windows 11 (64-bit)
- **Development Environment:** VS Code (Local Desktop)
- **Programming Language:** Python 3.x
- **Core Library:** OpenCV 4.x (Open Source Computer Vision Library)
- **Hardware Interface:** Integrated High-Definition (HD) Web Camera

3. Technical Architecture & Model Integration

The system utilizes a "Cascade of Classifiers" approach, where the output of one model serves as the input for the next.

Component	Model / Algorithm	Primary Purpose
Human Detection	HOG + Linear SVM	Identifies human silhouettes in a video frame.
Face Extraction	Region of Interest (ROI)	Isolates the face from the detected body bounding box.
Gender Classifier	Caffe Model (`gender_net`)	Binary classification (Male/Female) using CNN.

Age Estimator	Caffe Model (`age_net`)	Multi-class classification across 8 age brackets.

4. Execution, Results & Analysis

Scenario 1: Pedestrian Detection (HOG + SVM)

- **Execution:** The `HOGDescriptor_getDefaultPeopleDetector()` was initialized to scan the video frame for gradients matching the human form.
- **Outcome:** Successfully generated green bounding boxes around moving and stationary persons.
- **Analysis:** Using HOG+SVM provided a high frame rate on the CPU, though it is sensitive to partial occlusions (e.g., if a person is behind a desk).

Scenario 2: Demographic Analysis (DNN Caffe Models)

- **Execution:** The system extracted a "blob" from the detected face, resized it to 227×227 pixels, and performed mean subtraction for normalization.
- **Outcome:** The system accurately categorized individuals into brackets such as `(25-32)` or `(15-20)`.
- **Analysis:** The gender prediction was highly accurate, while age estimation showed a ± 1 bracket variance depending on lighting conditions and facial orientation.

Captured Data Sample:

Person ID: 1

Detection: [x:120, y:80, w:200, h:450]

Classification: Female, (25-32)

Confidence: 94.2%

5. Recommendations & Remediation

To transition this from a laboratory prototype to a production-ready security tool, the following enhancements are recommended:

1. **Switch to YOLO/SSD:** For high-density areas, replacing HOG with **YOLO (You Only Look Once)** would allow for simultaneous detection of dozens of people with higher accuracy.
2. **Lighting Normalization:** Implement **Histogram Equalization** before passing faces to the `age_net` to reduce errors caused by shadows.
3. **Optimization:** Use **OpenVINO** or **TensorRT** to speed up the inference time on local Intel or NVIDIA hardware.
4. **Privacy Masking:** For compliance (like GDPR), implement a "Blur" function on faces where the confidence score is below a certain threshold.

6. Key Learning Outcomes

- **Model Chaining:** Mastered the logic of passing data between different types of AI models (Traditional CV $\rightarrow$ Deep Learning).
- **Real-Time Data Streams:** Gained proficiency in handling `cv2.VideoCapture` and `cv2.waitKey` for smooth user-interface interaction.
- **Local Dependency Management:** Successfully configured Caffe `.prototxt` and `.caffemodel` files within a local VS Code directory, avoiding common path-related errors.